

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – DEBUGGING & OPTIMISATION TASK

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Debugging & Optimisation Task
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	A.DAITY RASHITHA	Reg. No.:	192311181
Section / BaDate	192311181	Date:	1.8.26

Instructions to Students

- Answer ALL debugging and optimisation tasks. Total Marks: 100.
- Carefully analyze the given program/code segment before identifying errors.
- Clearly identify logical, syntactical, and performance-related issues in the code.
- Apply systematic debugging techniques to correct the errors effectively.
- Optimize the given solution by improving efficiency, reducing unnecessary computations, or enhancing time complexity wherever possible.
- Provide proper justification for all corrections and optimization decisions.
- Write clean, structured, and well-documented code with appropriate comments.
- Maintain clarity, logical reasoning, and technical accuracy throughout the solution.

Question Type	Focus Area	Questions	Marks
Debugging & Optimisation Task	Identify errors, debug programs, and optimize algorithm efficiency using appropriate techniques	Q1 - Q2	100 (2 × 50)
GRAND TOTAL:		100 Marks	

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" and "DAA PROGRAMMING". Below this, there's a code editor with a Python function for binary search. The code has several errors highlighted in red: "high = mid - 1 # ERROR" and "return -1". To the right of the code editor, there's an "Input" field with the values "10 20 30 40 50" and "30", and an "Output" field. Below the code editor, there's a "Your Code" section with the following code:

```
1 def binary_search(arr, key):
2
3     low = 0
4     high = len(arr) - 1
5
6     while low <= high:
7
8         mid = (low + high) // 2
9
10
11         if arr[mid] == key:
12             return mid
13
14         elif arr[mid] > key:
15             high = mid - 1
16
17         else:
18             low = mid + 1
19
20
21     return -1
22
23 def binary_search(arr, key):
24
25     low = 0
26     high = len(arr) - 1
27
28     while low <= high:
29
30         mid = (low + high) // 2
```

At the bottom of the interface, there's a task description: "high = mid - 1 # ERROR" and "return -1".

SIMATS Online Lab
Ask Gemini
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=828

SIMATS ENGINEERING
SIMATS Online Programming Lab
DAA PROGRAMMING
ALLI DAITY RAKSHITHA 192311181RD

Your Code

```

1 def find_max(arr):
2
3     if not arr:
4         return None
5
6     max_val = arr[0]
7
8     for i in range(1, len(arr)):
9
10        if arr[i] > max_val:
11            max_val = arr[i]
12
13    return max_val
14
15 if __name__ == "__main__":
16     try:
17
18         user_input = input()
19
20         data = list(map(int, user_input.split()))
21
22         result = find_max(data)
23
24         if result is not None:
25             print(result)
26     except EOFError:
27         pass
28     except ValueError:
29         pass

```

Run
Save
Submit

Input

10 45 22 89 33

Output

29°C Partly cloudy
Search
15:26 26-08-2026

Code of Conduct

I K. A.DAITY RASHITHA(192311181) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____ DAITY RASHITHA ____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Marks Evaluation:

Question No.	Problem Identification (20%)	Debugging (25%)	Optimization (20%)	Analysis (20%)	Code Quality (15%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____